# Chapter 13. Classification of Covering Spaces

Equivalence of coverings, universal covering space, covering transformations, existence of coverings

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Throughout this chapter a covering map $p : E \to B$ is assumed to have $E$ and $B$ path connected and locally path connected. For $p(e_0) = b_0$ let $H_0 = p_*(\pi_1(E, e_0))$, a subgroup of $\pi_1(B, b_0)$ isomorphic to $\pi_1(E, e_0)$ (Theorem 9.17).

## 13.1 Equivalence of covering spaces

Covering maps $p : E \to B$ and $p' : E' \to B$ are **equivalent** if there is a homeomorphism $h : E \to E'$ with $p = p' \circ h$ (an **equivalence**).

**Lemma 13.1 (79.1, general lifting lemma)**

Let $p : E \to B$ be a covering map, $p(e_0) = b_0$, and $f : Y \to B$ continuous with $f(y_0) = b_0$, where $Y$ is path connected and locally path connected. Then $f$ has a lifting $\tilde{f} : Y \to E$ with $\tilde{f}(y_0) = e_0$ iff $f_*(\pi_1(Y, y_0)) \subset p_*(\pi_1(E, e_0))$. Such a lifting is unique.

**Proof.** If $\tilde{f}$ exists, $f_* = p_*\tilde{f}_*$.

Uniqueness: for $y_1 \in Y$ and a path $\alpha$ from $y_0$ to $y_1$, $\tilde{f} \circ \alpha$ is the lifting of $f \circ \alpha$ from $e_0$, so $\tilde{f}(y_1)$ is its end point.

Existence: define $\tilde{f}(y_1)$ as the end point of the lifting of $f \circ \alpha$ from $e_0$. It is well defined: if $\beta$ is another path from $y_0$ to $y_1$, lift $f \circ \alpha$ to $\gamma$ from $e_0$ and $f \circ \bar{\beta}$ to $\delta$ from $\gamma(1)$; then $\gamma * \delta$ lifts the loop $f \circ (\alpha * \bar{\beta})$, whose class lies in $f_*\pi_1(Y, y_0) \subset H_0$, so it is a loop (Theorem 9.17(c)); then $\bar{\delta}$ is the lifting of $f \circ \beta$ from $e_0$ and ends at $\gamma(1)$. Continuity at $y_1$: given a neighbourhood $N$ of $\tilde{f}(y_1)$, choose a path-connected evenly covered $U \ni f(y_1)$ whose slice $V_0$ through $\tilde{f}(y_1)$ lies in $N$, and a path-connected neighbourhood $W$ of $y_1$ with $f(W) \subset U$. For $y \in W$ and a path $\delta$ in $W$ from $y_1$ to $y$, the lifting of $f \circ (\alpha * \delta)$ is $\gamma * ((p|V_0)^{-1} \circ f \circ \delta)$, ending in $V_0$. So $\tilde{f}(W) \subset V_0 \subset N$. ■

**Theorem 13.2 (79.2)**

Let $p : E \to B$, $p' : E' \to B$ be covering maps with $p(e_0) = p'(e_0') = b_0$. There is an equivalence $h : E \to E'$ with $h(e_0) = e_0'$ iff $H_0 = p_*\pi_1(E, e_0)$ and $H_0' = p_*'\pi_1(E', e_0')$ are equal. Such $h$ is unique.

**Proof.** If $h$ exists, $h_*$ is an isomorphism and $p = p'h$ gives $H_0 = H_0'$. Conversely, Lemma 13.1 gives liftings $h : E \to E'$ of $p$ with $h(e_0) = e_0'$ and $k : E' \to E$ of $p'$ with $k(e_0') = e_0$. Then $k \circ h$ and $\mathrm{id}_E$ both lift $p$ and fix $e_0$, so they are equal by uniqueness; likewise $h \circ k = \mathrm{id}$. Uniqueness of $h$ is that of liftings. ■

**Lemma 13.3 (79.3)**

Let $e_0, e_1 \in p^{-1}(b_0)$ and $H_i = p_*\pi_1(E, e_i)$.
(a) If $\gamma$ is a path in $E$ from $e_0$ to $e_1$ and $\alpha = p \circ \gamma$, then $[\alpha]H_1[\alpha]^{-1} = H_0$.
(b) Conversely, if $H$ is conjugate to $H_0$, then $H = H_1$ for some $e_1 \in p^{-1}(b_0)$.

**Proof.** (a) If $[h] \in H_1$, $h = p \circ \tilde{h}$ with $\tilde{h}$ a loop at $e_1$; $\gamma * \tilde{h} * \bar{\gamma}$ is a loop at $e_0$ mapping to $[\alpha][h][\alpha]^{-1}$, so $[\alpha]H_1[\alpha]^{-1} \subset H_0$. Using $\bar{\gamma}$, $[\alpha]^{-1}H_0[\alpha] \subset H_1$.

(b) If $H_0 = [\alpha]H[\alpha]^{-1}$, lift $\alpha$ to $\gamma$ from $e_0$ and let $e_1 = \gamma(1)$; by (a), $H_0 = [\alpha]H_1[\alpha]^{-1}$, so $H = H_1$. ■

**Theorem 13.4 (79.4)**

$p$ and $p'$ are equivalent iff $H_0$ and $H_0'$ are conjugate in $\pi_1(B, b_0)$.

**Proof.** If $h$ is an equivalence and $e_1' = h(e_0)$, then $H_0 = p_*'\pi_1(E', e_1')$ by Theorem 13.2, which is conjugate to $H_0'$ by Lemma 13.3. Conversely, if they are conjugate, Lemma 13.3(b) gives $e_1'$ with $p_*'\pi_1(E', e_1') = H_0$, and Theorem 13.2 gives an equivalence carrying $e_0$ to $e_1'$. ■

**Example 13.1**

$\pi_1(S^1) \cong \mathbb{Z}$ is abelian with subgroups $n\mathbb{Z}$. So every covering of $S^1$ is equivalent to $p : \mathbb{R} \to S^1$ (trivial subgroup) or to $z \mapsto z^n$ (subgroup $n\mathbb{Z}$).

## 13.2 The universal covering space

If $E$ is simply connected, $p$ is a **universal covering**; by Theorem 13.4 it is unique up to equivalence.

**Lemma 13.5 (80.1)**

Let $B$ be path connected and locally path connected, and $p : E \to B$ a covering map with $E$ not necessarily path connected. If $E_0$ is a path component of $E$, then $p|E_0 : E_0 \to B$ is a covering map.

**Proof.** $E$ is locally path connected, so $E_0$ is open and $p(E_0)$ is open. If $x \in \overline{p(E_0)}$, take a path-connected evenly covered $U \ni x$; some slice $V$ meets $E_0$, and being path connected lies in $E_0$, so $x \in U = p(V) \subset p(E_0)$. Thus $p(E_0) = B$. For any such $U$, $(p|E_0)^{-1}(U)$ is the union of the slices meeting $E_0$, each contained in $E_0$ and mapped homeomorphically onto $U$. ■

**Lemma 13.6 (80.2)**

Let $p = r \circ q$ with $p : X \to Z$, $q : X \to Y$, $r : Y \to Z$ continuous (all spaces path connected and locally path connected).
(a) If $p$ and $r$ are covering maps, so is $q$.
(b) If $p$ and $q$ are covering maps, so is $r$.

**Proof.** Let $x_0 \in X$, $y_0 = q(x_0)$.

(a) $q$ is surjective: for a path $\tilde{\alpha}$ in $Y$ from $y_0$ to $y$, lift $r \circ \tilde{\alpha}$ through $p$ to $\tilde{\tilde{\alpha}}$ from $x_0$; $q \circ \tilde{\tilde{\alpha}}$ lifts $r \circ \tilde{\alpha}$ through $r$ from $y_0$, so equals $\tilde{\alpha}$, and $y$ is in the image. Given $y$, take a path-connected $U \ni r(y)$ evenly covered by both $p$ and $r$, and the slice $V$ of $r^{-1}(U)$ containing $y$. Each slice $U_\alpha$ of $p^{-1}(U)$ is connected, so $q$ maps it into one slice of $r^{-1}(U)$; $q^{-1}(V)$ is the union of those $U_\alpha$ mapped into $V$, and on each $q = (r|V)^{-1} \circ (p|U_\alpha)$ is a homeomorphism onto $V$.

(b) Let $U$ be a path-connected neighbourhood of $z$ evenly covered by $p$, and $V_\beta$ the path components of $r^{-1}(U)$, open and disjoint. The slices $U_\alpha$ of $p^{-1}(U)$ are the path components of $p^{-1}(U)$, and $q$ maps each into one $V_\beta$; so $q^{-1}(V_\beta)$ is a union of slices and, $q$ being surjective, contains some $U_\alpha$. By Theorem 9.10, $q$ restricts to a covering map $q^{-1}(V_\beta) \to V_\beta$, and $U_\alpha$ is a path component of $q^{-1}(V_\beta)$, so $q|U_\alpha : U_\alpha \to V_\beta$ is surjective by Lemma 13.5. It is also open and injective (as $p|U_\alpha$ is injective). Thus $q|U_\alpha$ is a homeomorphism onto $V_\beta$, and $r|V_\beta = (p|U_\alpha) \circ (q|U_\alpha)^{-1}$ is a homeomorphism onto $U$. ■

**Theorem 13.7 (80.3)**

If $p : E \to B$ is a covering map with $E$ simply connected, then for any covering map $r : Y \to B$ there is a covering map $q : E \to Y$ with $r \circ q = p$.

**Proof.** Lemma 13.1 applies trivially and gives $q$ with $r \circ q = p$; Lemma 13.6(a) shows $q$ is a covering map. ■

**Lemma 13.8 (80.4)**

If $p : E \to B$ is a covering map with $E$ simply connected and $p(e_0) = b_0$, then $b_0$ has a neighbourhood $U$ such that $i_* : \pi_1(U, b_0) \to \pi_1(B, b_0)$ is trivial.

**Proof.** Let $U$ be evenly covered and $\tilde{U}$ the slice containing $e_0$. A loop $f$ in $U$ lifts to a loop in $\tilde{U}$, which is nulhomotopic in $E$; applying $p$ gives a path homotopy in $B$ from $f$ to a constant. ■

**Example 13.2**

The infinite earring $X$ (Example 11.2) is path connected and locally path connected but has no universal covering: for any neighbourhood $U$ of the origin, some circle $C_n$ lies in $U$, and $C_n$ is a retract of $X$, so $\pi_1(C_n) \to \pi_1(X)$ is injective and $i_* : \pi_1(U) \to \pi_1(X)$ is nontrivial.

## 13.3 Covering transformations

An equivalence of $p$ with itself is a **covering transformation**; they form the group $\mathcal{C}(E, p, B)$. The **normalizer** of $H$ in $G$ is $N(H) = \{g : gHg^{-1} = H\}$, the largest subgroup in which $H$ is normal. Let $F = p^{-1}(b_0)$, $\Phi : \pi_1(B, b_0)/H_0 \to F$ the bijection induced by the lifting correspondence (Theorem 9.17), and $\Psi : \mathcal{C}(E, p, B) \to F$, $\Psi(h) = h(e_0)$, injective by uniqueness of liftings.

**Lemma 13.9 (81.1)**

The image of $\Psi$ equals $\Phi(N(H_0)/H_0)$.

**Proof.** Let $\gamma$ be the lifting of a loop $\alpha$ from $e_0$, $e_1 = \gamma(1) = \Phi([\alpha]H_0)$, $H_1 = p_*\pi_1(E, e_1)$. A covering transformation with $h(e_0) = e_1$ exists iff $H_1 = H_0$ (Theorem 13.2). Since $[\alpha]H_1[\alpha]^{-1} = H_0$ (Lemma 13.3), this holds iff $[\alpha]H_0[\alpha]^{-1} = H_0$, i.e. $[\alpha] \in N(H_0)$. ■

**Theorem 13.10 (81.2)**

$\Phi^{-1} \circ \Psi : \mathcal{C}(E, p, B) \to N(H_0)/H_0$ is a group isomorphism.

**Proof.** It is a bijection by Lemma 13.9. Let $h(e_0) = e_1$, $k(e_0) = e_2$, and choose paths $\gamma, \delta$ from $e_0$ to $e_1, e_2$; with $\alpha = p \circ \gamma$, $\beta = p \circ \delta$, $\Phi([\alpha]H_0) = e_1$ and $\Phi([\beta]H_0) = e_2$. Then $h \circ \delta$ runs from $e_1$ to $e_3 = h(k(e_0))$ and lifts $\beta$, so $\gamma * (h \circ \delta)$ lifts $\alpha * \beta$ from $e_0$ and ends at $e_3$: $\Phi([\alpha * \beta]H_0) = \Psi(h \circ k)$. ■

**Corollary 13.11 (81.3)**

$H_0$ is normal in $\pi_1(B, b_0)$ iff for all $e_1, e_2 \in p^{-1}(b_0)$ some covering transformation carries $e_1$ to $e_2$. Then $\mathcal{C}(E, p, B) \cong \pi_1(B, b_0)/H_0$.

**Proof.** If $H_0$ is normal, $\Psi$ is surjective, giving $h_i$ with $h_i(e_0) = e_i$; take $h_2 h_1^{-1}$. Conversely, transitivity makes $\Psi$ surjective, so $N(H_0)/H_0 = \pi_1/H_0$ and $N(H_0) = \pi_1(B, b_0)$. ■

**Corollary 13.12 (81.4)**

If $E$ is simply connected, $\mathcal{C}(E, p, B) \cong \pi_1(B, b_0)$.

If $H_0$ is normal, $p$ is **regular**. Then $\mathcal{C}$ is transitive on every fibre $p^{-1}(b)$: for $x_1, x_2 \in p^{-1}(b)$ and a path $\beta$ from $b$ to $b_0$, lift $\beta$ from $x_1, x_2$ to paths ending at $e_1, e_2$, and let $h(e_1) = e_2$; then $h$ carries the first lifting to a lifting of $\beta$ ending at $e_2$, which by uniqueness (lift $\bar{\beta}$ from $e_2$) is the second, so $h(x_1) = x_2$.

**Example 13.3**

Every covering of $S^1$ is regular; for $\mathbb{R} \to S^1$ the covering transformations are $x \mapsto x + n$. For the covering of the figure eight in Lemma 9.45, $\mathcal{C}$ is trivial: a loop around $A$ lifts to a non-loop from the origin but to a loop from every other point of the fibre on the $y$-axis, and similarly for $B$, so every covering transformation fixes the origin.

For a group $G$ of homeomorphisms of $X$, the **orbit space** $X/G$ is the quotient by $x \sim g(x)$. The action is **properly discontinuous** if each $x$ has a neighbourhood $U$ with $g(U) \cap U = \emptyset$ for $g \neq e$; then $g_0(U) \cap g_1(U) = \emptyset$ for $g_0 \neq g_1$.

**Theorem 13.13 (81.5)**

Let $X$ be path connected and locally path connected and $G$ a group of homeomorphisms of $X$. The quotient map $\pi : X \to X/G$ is a covering map iff the action is properly discontinuous. In that case $\pi$ is regular and $G$ is its group of covering transformations.

**Proof.** $\pi$ is open, since $\pi^{-1}\pi(U) = \bigcup_g g(U)$. If the action is properly discontinuous and $U$ is as in the definition, $\pi^{-1}\pi(U)$ is the disjoint union of the open sets $g(U)$, each meeting each orbit at most once; $\pi|g(U)$ is a continuous open bijection onto $\pi(U)$, hence a homeomorphism. Conversely, if $\pi$ is a covering, let $U$ be the slice through $x$ over an evenly covered neighbourhood of $\pi(x)$; if $g \neq e$ and $u' = g(u) \in U$ with $u \in U$, then $u' = u$ by injectivity of $\pi|U$; so $g$ fixes $u$, and $g$ is a covering transformation fixing a point, hence the identity (shown next), a contradiction. Each $g$ is a covering transformation. If $h$ is one with $h(x_1) = x_2$, then $x_1, x_2$ lie in one orbit, $g(x_1) = x_2$ for some $g$, and $h = g$ by Theorem 13.2. $G$ is transitive on fibres, so $\pi$ is regular (Corollary 13.11). ■

**Theorem 13.14 (81.6)**

If $p : X \to B$ is regular with group of covering transformations $G$, there is a homeomorphism $k : X/G \to B$ with $p = k \circ \pi$.

**Proof.** $p$ is constant on orbits, so it induces a continuous $k$. Conversely $p$ is a quotient map (continuous, open, surjective), and $\pi$ is constant on each fibre of $p$ since $G$ is transitive on fibres; so $\pi$ induces a continuous inverse $B \to X/G$. ■

**Example 13.4**

On the cylinder $X = S^1 \times I$, $h(x, t) = (-x, t)$ and $k(x, t) = (-x, 1 - t)$ generate properly discontinuous actions of $\mathbb{Z}/2$; $X/\langle h \rangle$ is again a cylinder, $X/\langle k \rangle$ is the Möbius band. $\mathbb{Z}^2$ acting on $\mathbb{R}^2$ by translations gives the torus; $\{\pm 1\}$ acting on $S^n$ gives $P^n$.

## 13.4 Existence of covering spaces

$B$ is **semilocally simply connected** if each $b$ has a neighbourhood $U$ such that $\pi_1(U,b) \to \pi_1(B,b)$ is trivial. By Lemma 13.8 this is necessary for a universal covering.

**Theorem 13.15 (82.1)**

Let $B$ be path connected, locally path connected and semilocally simply connected, $b_0 \in B$, and $H$ a subgroup of $\pi_1(B,b_0)$. There is a covering map $p : E \to B$ and $e_0 \in p^{-1}(b_0)$ with $p_*\pi_1(E,e_0) = H$.

**Proof.**

**Step 1.** Let $\mathcal{P}$ be the set of paths in $B$ from $b_0$. Put $\alpha \sim \beta$ if $\alpha(1) = \beta(1)$ and $[\alpha * \bar{\beta}] \in H$, an equivalence relation; write $\alpha^\#$ for the class, $E$ for the set of classes, and $p(\alpha^\#) = \alpha(1)$. Note:

(1) $[\alpha] = [\beta]$ implies $\alpha^\# = \beta^\#$;

(2) $\alpha^\# = \beta^\#$ implies $(\alpha * \delta)^\# = (\beta * \delta)^\#$ for any path $\delta$ from $\alpha(1)$, since $[(\alpha * \delta) * \overline{(\beta * \delta)}] = [\alpha * \bar{\beta}]$.

**Step 2.** For $\alpha \in \mathcal{P}$ and a path-connected neighbourhood $U$ of $\alpha(1)$ let $B(U,\alpha) = \{(\alpha * \delta)^\# : \delta$ a path in $U$ from $\alpha(1)\}$; it contains $\alpha^\#$. If $\beta^\# \in B(U,\alpha)$, then $B(U,\beta) = B(U,\alpha)$: with $\beta^\# = (\alpha * \delta)^\#$, elements $(\beta * \gamma)^\# = (\alpha * (\delta * \gamma))^\#$ lie in $B(U,\alpha)$ by (2), and $\alpha^\# = (\beta * \bar{\delta})^\#$ gives the reverse inclusion. Hence if $\beta^\# \in B(U_1,\alpha_1) \cap B(U_2,\alpha_2)$, choosing a path-connected $V \subset U_1 \cap U_2$ containing $\beta(1)$, $B(V,\beta) \subset B(U_1,\beta) \cap B(U_2,\beta) = B(U_1,\alpha_1) \cap B(U_2,\alpha_2)$. So these sets form a basis.

**Step 3.** $p$ is continuous ($B(U,\alpha) \subset p^{-1}(W)$ when $U \subset W$) and open ($p(B(U,\alpha)) = U$).

**Step 4.** Let $U$ be a path-connected neighbourhood of $b_1$ with $\pi_1(U,b_1) \to \pi_1(B,b_1)$ trivial. Then $p^{-1}(U)$ is the union of the sets $B(U,\alpha)$ with $\alpha(1) = b_1$: if $p(\beta^\#) \in U$, choose $\delta$ in $U$ from $b_1$ to $\beta(1)$ and $\alpha = \beta * \bar{\delta}$; then $\beta^\# = (\alpha * \delta)^\#$. Distinct such sets are disjoint by Step 2. And $p$ is injective on $B(U,\alpha)$: if $(\alpha * \delta_1)(1) = (\alpha * \delta_2)(1)$, then $\delta_1 * \bar{\delta}_2$ is a loop in $U$ at $b_1$, nulhomotopic in $B$, so $[\alpha * \delta_1] = [\alpha * \delta_2]$. Being a continuous open bijection, $p|B(U,\alpha)$ is a homeomorphism onto $U$.

**Step 5.** Let $e_0$ be the class of the constant path. For $\alpha \in \mathcal{P}$ and $c \in I$ let $\alpha_c(t) = \alpha(tc)$, and $\tilde{\alpha}(c) = \alpha_c^\#$. It lifts $\alpha$ from $e_0$ to $\alpha^\#$. It is continuous at $c$: a basic neighbourhood of $\tilde{\alpha}(c)$ is $B(U,\alpha_c)$, $U$ a path-connected neighbourhood of $\alpha(c)$; for $d$ near $c$, $\alpha([c,d]) \subset U$, and $[\alpha_d] = [\alpha_c * \delta]$ with $\delta$ the reparametrized restriction of $\alpha$ to $[c,d]$ (or $[d,c]$, reversed), so $\tilde{\alpha}(d) \in B(U,\alpha_c)$.

**Step 6.** $E$ is path connected by Step 5, and locally path connected since it is locally homeomorphic to $B$; so $p$ is a covering map.

**Step 7.** For a loop $\alpha$ at $b_0$, its lifting from $e_0$ ends at $\alpha^\#$, which equals $e_0$ iff $[\alpha] \in H$. By Theorem 9.17(c), $p_*\pi_1(E,e_0) = H$. ∎

**Corollary 13.16 (82.2)**

$B$ has a universal covering space iff it is path connected, locally path connected and semilocally simply connected.

**Proof.** Necessity: Lemma 13.8 (each point of $B$ is in the image of a point of $E$).

Sufficiency: Theorem 13.15 with $H$ trivial. ∎

Together with Theorem 13.4: for such $B$, equivalence classes of coverings correspond bijectively to conjugacy classes of subgroups of $\pi_1(B,b_0)$.